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Capstone Project Report

Dual-View RAG-VLM: Baseline Auditing, Classifier-Conditioned Generation and Consensus Decoding for Chest X-Ray Report Generation Under Consumer Hardware Constraints

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Abstract

Automated radiology report generation can help reduce diagnostic backlog and radiologist workload. However, reported progress on the Indiana University Chest X-Ray (IU X-Ray) benchmark is difficult to interpret, because reports are highly repetitive and published results are rarely compared against trivial baselines or obtained under a common evaluation protocol. This work presents **Dual-View RAG-VLM**, a parameter-efficient dual-projection vision-language model trained and evaluated on consumer-grade GPU hardware (NVIDIA RTX 4050, 6 GB VRAM). Frontal and lateral radiographs are encoded by a frozen RAD-DINO encoder. The 768-dimensional frontal and lateral vectors are concatenated into a 1,536-dimensional representation that feeds a logistic-regression classifier over 14 CheXpert findings and a FAISS index built only from training studies. Pooled visual tokens (288 per study), out-of-fold classifier probabilities, and two retrieved training reports are passed to Qwen2.5-3B-Instruct, which is adapted with 4-bit QLoRA. At inference, a pool of candidate reports is generated and the final report is selected by Minimum Bayes Risk (MBR) consensus decoding, which serves as the verification stage of the pipeline.

On the locked test set of 589 studies, a constant normal report (E0) reaches BLEU-1 of 0.2390 and ROUGE-L of 0.2809, and copying the top-1 retrieved training report (E1) reaches BLEU-1 of 0.3449 and METEOR of 0.1514. Beam search with the fine-tuned model falls below the retrieval baseline on BLEU-1 (0.2728) because it tends to collapse to generic templates, whereas MBR consensus decoding with eight candidates reaches BLEU-1 of 0.3820, METEOR of 0.1816, and CIDEr of 0.3240 without changing model weights. The final implementation enlarges the pool to sixteen candidates; in a 50-study pilot it reaches BLEU-1 of 0.4053, METEOR of 0.1926, and ROUGE-L of 0.3124, but its evaluation on the full test set is left as future work. A linear probe on the frozen dual-view representations achieves a macro-AUC of 0.8135 over 14 CheXpert findings. These results are obtained without large-scale pre-training on MIMIC-CXR, remain below published state-of-the-art scores, and are not directly comparable with them.

Keywords: Chest X-Ray, Radiology Report Generation, Dual-View Learning, RAD-DINO, Retrieval-Augmented Generation, Qwen2.5, QLoRA, Minimum Bayes Risk Decoding, Baseline Auditing, Medical Vision-Language Models

1 Introduction

Chest X-ray imaging is one of the most widely used diagnostic imaging modalities in clinical practice. Radiologists examine multi-view radiographs, typically a frontal and a lateral projection, identify abnormalities, and summarize their observations in a textual

report. The growing volume of imaging studies and the shortage of radiologists have increased interest in automatic radiology report generation, which could reduce reporting time and support clinical workflows by converting visual findings into structured clinical narratives [11, 14].

Recent research has explored knowledge-enhanced multimodal attention, self-supervised learning, Transformer-based architectures, contrastive learning, expert knowledge retrieval, cross-modal interaction, and reinforcement learning to improve automatic radiology report generation. Many of these systems report strong n-gram scores on the IU X-Ray collection [12]; for example, DART and KiUT report BLEU-1 values of roughly 0.49–0.53 [15, 16]. However, these numbers need to be interpreted with care for three reasons:

1. **Pre-training confounding:** Many top-performing systems are pre-trained on large paired corpora such as MIMIC-CXR [13] using multi-GPU clusters before evaluation on IU X-Ray.
2. **Template redundancy and missing baselines:** Routine chest radiographs are frequently normal, so reports share extensive wording. Standard n-gram metrics reward this repetition, yet few studies report how much score is achieved by a constant normal report or by simple nearest-neighbor retrieval.
3. **Hardware accessibility:** Increasingly large multimodal models exclude academic laboratories and student researchers who do not have large compute budgets.

To address these issues, this work proposes the **Dual-View RAG-VLM** and evaluates it under a fixed, leakage-free protocol on consumer hardware. The frontal and lateral radiographs are encoded by a frozen RAD-DINO encoder. A logistic-regression classifier over the concatenated 1,536-dimensional representation provides out-of-fold probabilistic priors, a FAISS index restricted to training studies provides similar reports, and a two-layer MLP projector maps the visual tokens into the embedding space of Qwen2.5-3B-Instruct, which is adapted with 4-bit QLoRA. At inference, several candidate reports are generated and the final report is selected by MBR consensus decoding.

The main contributions of this work are as follows:

- **Empirical baseline floors:** A constant normal report (E0) and a pure top-1 retrieval baseline (E1) are evaluated on the exact test set ($N = 589$), establishing the minimum performance that a learned generator should exceed.
- **Dual-view multimodal architecture:** Frozen RAD-DINO tokens pooled to 288 dual-view tokens, learned view embeddings, a logistic-regression finding classifier whose out-of-fold probabilities condition the prompt, train-only retrieval,

and QLoRA fine-tuning of Qwen2.5-3B-Instruct, trainable within about 4.9 GB of VRAM.

- **Clinical representation check:** A linear probe over 14 CheXpert categories shows that the frozen dual-view features carry useful pathological signal (macro-AUC 0.8135).
- **Decoding strategy comparison:** Beam search, stochastic sampling, and MBR consensus decoding are compared on the same model; MBR with eight candidates substantially outperforms beam search on BLEU-1 and METEOR without changing the model weights.
- **Failure analysis:** A qualitative analysis shows both what consensus decoding fixes (unsupported statements that appear in single rollouts) and what it does not (abnormal studies that still receive normal-template reports).

2 Literature Review & Table

Automatic radiology report generation has evolved from conventional image-to-text approaches toward multimodal systems that incorporate medical knowledge, attention mechanisms, contrastive learning, retrieval, and large vision-language models. Recent studies have primarily focused on improving the quality, relevance, and clinical content of automatically generated reports.

Xu et al. (2023) proposed *Radiology Report Generation via Structured Knowledge-Enhanced Multi-modal Attention and Contrastive Learning*, which combines structured medical knowledge with multimodal attention and dual-branch contrastive learning. Although knowledge fusion improves report generation, the approach does not provide a dedicated mechanism for determining whether generated clinical claims are supported by the input image [1].

Pan et al. (2024) introduced S3-Net, a self-supervised dual-stream network that combines CNN and Swin Transformer representations with a Fusion Alignment Module. The approach improves visual representation and visual-text alignment, but its primary focus remains report generation rather than independent verification of generated findings [2].

Yi et al. (2024) proposed the Two-Stage Global Enhanced Transformer (TSGET), which connects global image features with generated words and subsequently incorporates region-level visual information. While the model addresses global and regional visual representation, it primarily optimizes report-generation performance rather than detecting unsupported clinical statements [3].

Liu et al. (2024) proposed *Multi-Grained Radiology Report Generation with Sentence-Level Image-Language Contrastive Learning*. The method learns sentence-level topics and fine-grained image representations using contrastive learning and employs multiple decoders for report generation. The study identifies challenges involving rare findings, subtle abnormalities, uncertainty, and long or complex sentences [4].

Zhang et al. (2024) developed a Visual-Textual Cross-Modal Interaction Network to address the problem of abnormal regions occupying only a small portion of chest X-ray images. The model improves visual-textual interaction and abnormality representation but does not introduce a post-generation verification mechanism [5].

Bu et al. (2024) proposed EKAGen, which incorporates instance-level expert knowledge, discriminative attention, and self-distillation to improve disease-specific visual representation. Although the approach makes better use of expert knowledge and important image regions, it does not independently verify each generated clinical claim [6].

Gu et al. (2024) introduced an Automatic Medical Report Generation approach based on a Detector Attention Module and GPT-Based Word LSTM. The detector attention module provides lesion-location information while GPT-based semantic information contributes to report generation. However, the approach does not explicitly determine whether generated statements are clinically supported [7].

Hou et al. (2024) proposed Retrieval Enhanced Cross-modal Fusion for radiographic report generation. Retrieval is used to provide relevant information during generation, demonstrating the potential of retrieval-based contextual enhancement. However, retrieval alone does not guarantee that the final generated report is faithful to the current X-ray image [8].

Liao et al. (2024) investigated Information Extraction to Normalize Training Data for Automatic Radiology Report Generation, using clinical entity and relation extraction to transform narrative reports into structured representations. The study also highlights limitations related to clinical context, uncertain observations, and generalization across different clinical sites [9].

Yi et al. (2025) proposed LHR-RFL, a linear hybrid-reward-based reinforced focal learning approach that gives greater importance to difficult training cases. The work focuses on improving report-generation performance on difficult samples but does not introduce a dedicated verification layer for hallucinated findings [10].

2.1 Literature Review Table

Table 1 summarizes the ten reviewed studies, their major approaches, contributions, and limitations relevant to the proposed work.

Table 1: Summary of the Reviewed Literature

Paper / Year	**Method**	**Key Contribution**	**Relevant Limitation**
Xu et al. (2023)	Knowledge-enhanced multi-modal attention and contrastive learning	Combines structured medical knowledge with multi-modal attention and dual-branch contrastive learning for radiology report generation.	Does not provide a dedicated mechanism for verifying whether generated clinical claims are supported by the input image.
Pan et al. (2024)	S3-Net; self-supervised dual-stream CNN and Swin Transformer	Uses dual-stream visual representations and a Fusion Alignment Module to improve visual-text alignment.	Focuses primarily on report generation and does not independently verify generated findings.
Yi et al. (2024)	Two-Stage Global Enhanced Transformer (TS-GET)	Combines global image features with generated words and incorporates region-level visual information.	Primarily optimizes report generation rather than detecting unsupported clinical statements.
Liu et al. (2024)	Multi-grained report generation with sentence-level image-language contrastive learning	Learns sentence-level topics and fine-grained image representations using contrastive learning.	Challenges remain for rare findings, subtle abnormalities, uncertainty, and complex reports.
Zhang et al. (2024)	Visual-Textual Cross-Modal Interaction Network	Improves visual-textual interaction for abnormal regions occupying relatively small image areas.	Does not introduce a dedicated post-generation verification mechanism.
Bu et al. (2024)	EKAGen; expert knowledge-aware report generation	Uses instance-level expert knowledge, discriminative attention, and self-distillation for disease-specific representation.	Does not independently verify each generated clinical claim.

Paper / Year	Method	Key Contribution	Relevant Limitation
Gu et al. (2024)	Detector Attention Module and GPT-Based Word LSTM	Combines lesion-location information from detector attention with GPT-based semantic information for report generation.	Does not explicitly determine whether generated statements are clinically supported.
Hou et al. (2024)	Retrieval Enhanced Cross-modal Fusion	Uses retrieval to provide relevant information during radiographic report generation.	Retrieved information alone does not guarantee faithfulness to the current X-ray image.
Liao et al. (2024)	Information extraction for training-data normalization	Uses clinical entity and relation extraction to transform narrative reports into structured representations.	Highlights limitations related to clinical context, uncertain observations, and cross-site generalization.
Yi et al. (2025)	LHR-RFL; hybrid-reward reinforced focal learning	Emphasizes difficult training cases to improve radiology report-generation performance.	Does not introduce a dedicated verification layer for hallucinated findings.

2.2 Research Gap

The reviewed literature demonstrates substantial progress in automatic chest X-ray report generation through improved visual representation, multimodal alignment, medical knowledge integration, retrieval, and training strategies. However, the reviewed approaches primarily concentrate on generation quality and typically report n-gram scores without calibrating them against trivial domain baselines. Many also rely on large-scale pre-training, use a single decoding pass, and are evaluated under protocols that differ in splits, text preprocessing, and scoring tools. According to the literature survey conducted for this work, few studies jointly (i) audit the benchmark with constant-report and pure-retrieval baselines, (ii) exploit both radiograph views within a compact model that fits on consumer hardware, and (iii) examine how the decoding strategy affects the tendency of vision-language models to collapse to generic templates.

This motivates the development of a dual-view, retrieval-augmented, parameter-efficient framework with consensus decoding, evaluated under a fixed and leakage-free protocol

against explicit empirical baselines.

3 Problem Statement and Objectives

3.1 Problem Statement

Automated chest X-ray report generation systems have demonstrated the ability to convert medical images into textual radiology reports. However, many of these systems use a single radiograph view, depend on image features without clinically grounded context, and select the final report from a single decoding pass. Because reports in the benchmark dataset are highly repetitive, n-gram metrics can be inflated by generic wording, and reported improvements are difficult to interpret without baselines. Retrieval components and clinical indicators can also introduce train–test mismatch or information leakage, and the compute requirements of large models limit reproducibility by smaller research groups.

Therefore, there is a need for a chest X-ray report generation framework that exploits both frontal and lateral views, grounds generation in clinical priors and leakage-free retrieved reports, selects reports through a consensus-based decoding strategy, runs on consumer hardware, and is evaluated against explicit baselines under a fixed and reproducible protocol.

3.2 Objectives

The main objectives of the proposed work are:

1. To build a dual-view (frontal and lateral) visual encoding pipeline using a frozen, radiology-specific RAD-DINO encoder with spatial pooling and learned view embeddings.

2. To train a logistic-regression image classifier over 14 CheXpert findings on the concatenated 1,536-dimensional representation and use its out-of-fold probabilities as clinical priors that are available identically during training and inference.

3. To incorporate leakage-free retrieval-augmented generation, in which similar training reports are retrieved from a FAISS index built exclusively from the training split.

4. To adapt Qwen2.5-3B-Instruct to radiology findings generation using 4-bit NF4 quantization and LoRA (QLoRA) within the memory limits of a consumer GPU.

5. To compare beam search, stochastic sampling, and Minimum Bayes Risk (MBR) consensus decoding, and to select the final report from a candidate pool by consensus.

6. To establish empirical baselines (constant normal report and pure retrieval) and evaluate all systems under a fixed protocol with the official `pycocoevalcap` toolkit.

4 Existing System and Proposed System

4.1 Existing System

Existing automated chest X-ray report generation systems generally follow an image-to-text generation paradigm. A deep learning model extracts visual features from a chest X-ray and a language-generation component converts these features into a textual radiology report.

Recent systems have incorporated attention mechanisms, Transformer architectures, contrastive learning, medical knowledge, retrieval, and reinforcement learning to improve report quality. Many of them are pre-trained on large report corpora before being evaluated on IU X-Ray, and they usually decode reports with a single beam-search or sampling pass.

4.2 Limitations of Existing Systems

The major limitations considered in this work are:

- **Single-view input:** Many systems use only the frontal radiograph and ignore the lateral view, which can contain complementary evidence.
- **Missing baselines:** Reported n-gram scores are rarely compared against a constant normal report or a pure retrieval baseline, although reports are highly repetitive.
- **Template mode collapse:** Likelihood-maximizing decoding such as beam search favors safe, high-frequency normal phrasing over less frequent, specific findings.
- **Retrieval leakage and prior mismatch:** Retrieval over a pool that includes evaluation studies, or clinical indicators that differ in reliability between training and inference, can inflate scores or mislead the model.

- **Computational cost:** Large pre-training corpora and large models restrict reproducibility for researchers without large compute budgets.
- **Inconsistent evaluation:** Differences in splits, truncation, casing, and scoring tools make fair comparison with published results difficult.

4.3 Proposed System

The proposed system, referred to as the *Dual-View RAG-VLM*, addresses these limitations through a pipeline with three parts: an image classifier block, an LLM block, and a consensus verification block.

Image classifier block. The frontal and lateral radiographs of each study are encoded by a frozen RAD-DINO encoder. The 768-dimensional frontal vector and the 768-dimensional lateral vector are concatenated into a 1,536-dimensional representation. A logistic-regression classifier maps this representation to 14 CheXpert finding probabilities, which are out-of-fold for training studies. The same 1,536-dimensional vector queries a FAISS index built only from training studies, which returns the two most similar training reports.

LLM block. The patch features of each view are pooled into 144 visual tokens, and learned view embeddings are added to distinguish the two views. A two-layer MLP projector maps the 288 visual tokens into the embedding space of Qwen2.5-3B-Instruct, which is fine-tuned using QLoRA. The visual tokens, finding probabilities, and retrieved reports are combined in a ChatML-style prompt. The model generates a pool of candidate reports consisting of beam-search outputs and stochastic samples.

Consensus verification block. Instead of keeping a single decoded output, the system selects the candidate that agrees most with the rest of the pool using Minimum Bayes Risk (MBR) decoding. Statements that appear in only one rollout are thereby disfavored. In the current implementation, the classifier output influences the report through the prompt; no separate post-hoc rewriting of the selected report is applied.

Two zero-parameter baselines, a constant normal report and a pure retrieval baseline, are evaluated under the same protocol.

4.4 Advantages of the Proposed System

The proposed framework provides the following capabilities:

- Dual-view (frontal and lateral) visual evidence with a frozen radiology-specific encoder.

- Training and inference on consumer-grade GPU hardware.
- Out-of-fold clinical priors that behave identically at training and inference time.
- Leakage-free retrieval restricted to the training split.
- Parameter-efficient adaptation of a 3B-parameter language model with 4-bit QLoRA.
- Consensus-based report selection that mitigates template mode collapse.
- Explicit empirical baselines and a fixed, documented evaluation protocol.

5 Data Set Description

5.1 IU Chest X-Ray Dataset

The proposed system uses the IU Chest X-Ray (IU X-Ray) collection, associated with Indiana University and the Open-i biomedical image search system [12]. The dataset contains chest X-ray images together with corresponding radiology reports, making it suitable for multimodal chest X-ray analysis and automatic radiology report generation.

The raw dataset contains 3,851 reports and 7,466 X-ray images. The image and report information is distributed across two principal CSV files, namely `indiana_projections.csv` and `indiana_reports.csv`, which are linked through a common `uid` serving as the study identifier. The projections file contains the fields `uid`, `filename`, and `projection`; among the images, 3,818 are frontal and 3,648 are lateral. The reports file contains the fields `uid`, `MeSH`, `Problems`, `image`, `indication`, `comparison`, `findings`, and `impression`. The `findings` field contains the radiologist observations and serves as the target text for report generation.

5.2 Study Selection and Data Splits

The evaluation follows a dual-projection protocol. A study is retained only if it contains (i) at least one frontal image, (ii) at least one lateral image, and (iii) a non-empty `findings` field. Single-view studies (174 cases) and records with empty findings are excluded, which leaves 2,943 usable studies.

The usable studies are partitioned at the study (`uid`) level into 2,060 training, 294 validation, and 589 test studies using a fixed random seed, and the split identifiers are locked. Because the available CSV files do not contain a patient identifier, patient-level isolation cannot be enforced, and this is acknowledged as a limitation.

5.3 Text Preprocessing

References and model outputs are lowercased and truncated to 60 words, following the convention of Chen et al. [11]. Anonymization placeholders are removed using a word-boundary substitution that matches two or more consecutive "x" characters (`re.sub(r'\bx{2,}\b', ' ', text)`). This avoids the substring-deletion error of character-level removal, which would corrupt clinical words such as "pneumothorax".

5.4 Dataset Statistics

Table 2: Summary of the IU Chest X-Ray Dataset and the Adopted Protocol

Property	**Value**
Dataset	IU Chest X-Ray
Total reports (raw)	3,851
Total X-ray images (raw)	7,466
Frontal images (raw)	3,818
Lateral images (raw)	3,648
Single-view studies excluded	174
Usable dual-view studies with findings	2,943
Training studies	2,060
Validation studies	294
Test studies	589
Auxiliary finding labels	14 (CheXpert categories)
Split level	Study (`uid`)
Maximum report length	60 words

6 Proposed System Architecture

6.1 Overall Architecture

Figure 1 presents the overall architecture. Both views are encoded by a frozen RAD-DINO encoder [17], and the resulting patch-token grids are spatially pooled into 144 tokens per view. Learned view embeddings are added to distinguish frontal from lateral tokens, giving 288 visual tokens per study.

The mean-pooled 768-dimensional features of the two views are concatenated into a 1,536-dimensional global vector. This vector serves two roles: it is the input to the logistic-regression finding classifier, and it is the query to a FAISS index [22] that returns the most similar training reports. In parallel, the visual tokens (the "image features" in Figure 1) are projected into the language-model embedding space by an MLP projector.

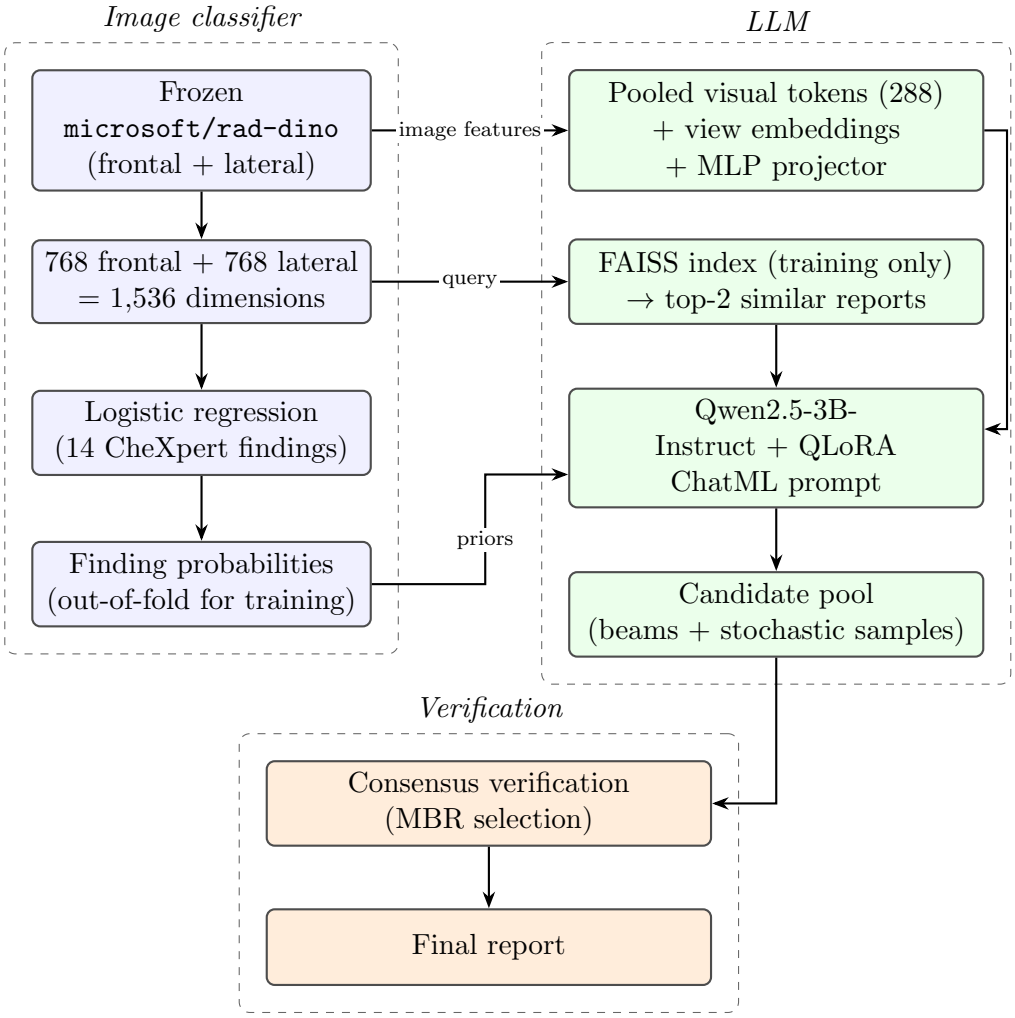


Figure 1: Overall architecture of the proposed Dual-View RAG-VLM. A frozen RAD-DINO encoder feeds (i) a logistic-regression image classifier over the 1,536-dimensional dual-view vector, (ii) a training-only FAISS retriever, and (iii) a projector into the Qwen2.5-3B-Instruct model fine-tuned with QLoRA. The classifier probabilities and retrieved reports enter the prompt. A pool of candidate reports is generated, and the final report is selected by MBR consensus.

The projected visual tokens, the predicted finding probabilities, and the retrieved reports are assembled into a ChatML-style prompt and passed to Qwen2.5-3B-Instruct [18] with QLoRA adapters [19]. A pool of candidate reports is generated, and MBR consensus decoding [21] selects the final report.

6.2 Architecture Components

6.2.1 Input Layer

The system receives the frontal and lateral radiographs of a study. Studies lacking either view are excluded from the primary protocol.

6.2.2 Dual-View Visual Encoder

A frozen `microsoft/rad-dino` encoder (ViT-B/14, pre-trained with self-supervised objectives on radiographs) extracts patch-level features from each view at a native resolution of 518×518 pixels. Its outputs are computed once and cached.

6.2.3 Visual Token Pooling and View Embeddings

Each view yields $37 \times 37 = 1{,}369$ patch tokens of dimension 768. Adaptive average pooling reduces each view to a 12×12 grid (144 tokens), and a learned view embedding is added to the tokens of each view. The two sets are concatenated, giving 288 visual tokens.

6.2.4 Image Classifier

The 768-dimensional frontal vector and the 768-dimensional lateral vector are concatenated into a 1,536-dimensional representation. A logistic-regression classifier on this representation predicts 14 CheXpert findings. Its probabilities are used as clinical priors in the prompt. For training studies the probabilities are 5-fold out-of-fold predictions.

6.2.5 Leakage-Free Retrieval Module

A FAISS `IndexFlatIP` index is built over L2-normalized global embeddings of the 2,060 training studies only. For a query study, the two most similar training reports are retrieved and included in the prompt.

6.2.6 MLP Projector

A two-layer MLP with GELU activation maps encoder features (768-dimensional) to the hidden size of the language model (2,048-dimensional).

6.2.7 Language Model with QLoRA

Qwen2.5-3B-Instruct is loaded with 4-bit NF4 quantization and adapted using LoRA on all attention and MLP projection layers. It generates the findings section conditioned on the multimodal prompt.

6.2.8 Consensus Decoding Module

A pool of candidate reports is generated, and the candidate with the highest average agreement with the others is selected as the final report.

6.2.9 Final Output

The final output is the generated findings section of the radiology report for the input study.

7 Technologies Used

The major technologies used in the proposed system are summarized in Table 3.

Table 3: Technologies Used in the Proposed System

Category	**Technology**
Programming Language	Python
Deep Learning Framework	PyTorch 2.7.0 (CUDA 12.8), bfloat16
Visual Encoder	RAD-DINO (`microsoft/rad-dino`), frozen
Image Classifier	Logistic regression over 1,536-d dual-view vector (14 findings)
Retrieval	FAISS (`IndexFlatIP`)
Language Model	Qwen2.5-3B-Instruct
Parameter-Efficient Fine-Tuning	LoRA (QLoRA)
Model Quantization	4-bit NF4 with double quantization
Transformer Framework	Hugging Face Transformers
Decoding	Beam search, nucleus sampling, MBR consensus
Evaluation	`pycocoevalcap` with Stanford PTBTokenizer (Java Runtime)
Data Processing	NumPy, Pandas
Visualization	Matplotlib
Hardware	NVIDIA GeForce RTX 4050 Laptop GPU (6 GB VRAM)

8 Methodology

The proposed framework consists of a sequential pipeline comprising data preparation, dual-view visual encoding, image classification with out-of-fold priors, leakage-free retrieval, multimodal prompt construction, two-stage QLoRA fine-tuning, consensus decoding, and baseline-based evaluation.

8.1 Dual-View Feature Extraction and Spatial Pooling

Rather than general-domain vision models trained on photographs, a frozen `microsoft/rad-dino` encoder [17] is used. For the frontal image $\mathbf{I}_{\text{front}}$ and the lateral image $\mathbf{I}_{\text{lat}}$, the encoder outputs $37 \times 37 = 1{,}369$ patch tokens of dimension $D_{\text{enc}} = 768$. Passing $2 \times 1{,}369 = 2{,}738$ visual tokens to a language model would exceed the memory and context limits of a 6 GB GPU, so adaptive 2D average pooling is applied:

$$\text{Pool}: \ \mathbb{R}^{37\times37\times768} \longrightarrow \mathbb{R}^{12\times12\times768}, \tag{1}$$

which reduces each view to $K = 144$ tokens. To preserve the projection identity, learned view embeddings $\mathbf{E}_{\text{front}}, \mathbf{E}_{\text{lat}} \in \mathbb{R}^{1\times144\times768}$ are added:

$$\mathbf{Z}_{\text{front}} = \text{Pool}(\mathbf{H}_{\text{front}}) + \mathbf{E}_{\text{front}}, \qquad \mathbf{Z}_{\text{lat}} = \text{Pool}(\mathbf{H}_{\text{lat}}) + \mathbf{E}_{\text{lat}}. \tag{2}$$

The two views are concatenated along the sequence dimension:

$$\mathbf{Z}_{\text{vis}} = [\mathbf{Z}_{\text{front}}\,;\,\mathbf{Z}_{\text{lat}}] \in \mathbb{R}^{1\times288\times768}. \tag{3}$$

Because the encoder is frozen, its outputs are cached and the encoder does not take part in the training loop.

8.2 Image Classifier and Clinical Priors

A 1,536-dimensional global vector is formed by concatenating the mean-pooled frontal and lateral features, $\mathbf{g} = [\text{mean}(\mathbf{H}_{\text{front}})\,;\,\text{mean}(\mathbf{H}_{\text{lat}})]$. A multi-label logistic-regression classifier over the 14 CheXpert findings [23] produces

$$\hat{\mathbf{p}} = \sigma(\mathbf{W}_{\text{cls}}\mathbf{g} + \mathbf{b}_{\text{cls}}) \in [0,1]^{14}. \tag{4}$$

The prompt probabilities are never ground-truth labels. For training studies they are 5-fold out-of-fold predictions, and for validation and test studies they come from a classifier fitted on the training set, so that the language model sees priors of comparable reliability during training and inference. The probabilities are written into the user prompt as a list of predicted indicators.

8.3 Multimodal Projector and Training Objective

The visual tokens are projected into the language-model hidden dimension ($D_{\text{llm}} = 2048$) by a two-layer MLP with GELU activation:

$$\mathbf{V}_{\text{tokens}} = \mathbf{W}_2\,\text{GELU}(\mathbf{W}_1\mathbf{Z}_{\text{vis}} + \mathbf{b}_1) + \mathbf{b}_2. \tag{5}$$

An auxiliary linear classification head on $\mathbf{g}$ is trained jointly with the report objective. The training objective combines the report cross-entropy and the auxiliary classification loss:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{CE}}(\mathbf{y}_{\text{tokens}}, \hat{\mathbf{y}}_{\text{tokens}}) + \lambda_{\text{cls}}\,\mathcal{L}_{\text{BCE}}(\mathbf{y}_{14}, \hat{\mathbf{p}}_{\text{head}}), \qquad \lambda_{\text{cls}} = 0.20. \tag{6}$$

8.4 Leakage-Free Dynamic Retrieval

For each study, the $k = 2$ most visually similar training cases are retrieved. To prevent leakage and trivial memorization during training, the query study is excluded from its own neighborhood:

$$\mathcal{N}(j) = \text{Top-}k\Big(\{\, i \in \mathcal{D}_{\text{train}} \setminus \{j\} \mid \langle \mathbf{g}_j, \mathbf{g}_i \rangle \,\} \Big). \tag{7}$$

Validation and test studies are never indexed. The retrieved reports and the finding probabilities are placed in a ChatML user prompt that contains 288 `<|image_pad|>` positions, which are replaced by the projected visual embeddings. The system message asks the model to act as an expert radiologist, write the findings section, and use only what the images and indicators support. The loss is computed only on the assistant tokens.

8.5 Two-Stage Training with QLoRA

Qwen2.5-3B-Instruct [18] serves as the language model. Its base weights are frozen in 4-bit NormalFloat (NF4) with double quantization [19]. Training has two stages. In Stage 1 (warm-up), only the projector and classification head are trained so that the visual tokens are aligned with the language-model embedding space. In Stage 2 (supervised fine-tuning), LoRA adapters [20] with $r = 32$ and $\alpha = 64$ are trained together with the projector and the classification head, and the checkpoint with the best validation performance is kept. LoRA is applied to all attention projections (`q_proj`, `k_proj`, `v_proj`, `o_proj`) and MLP layers (`gate_proj`, `up_proj`, `down_proj`). For a frozen weight matrix W_0, the adapted weight is

$$W = W_0 + \frac{\alpha}{r} BA, \qquad B \in \mathbb{R}^{d \times r},\ A \in \mathbb{R}^{r \times k}. \tag{8}$$

This gives 59,867,136 trainable parameters, about 1.90% of the 3.14-billion-parameter model, and allows forward and backward passes within about 4.9 GB of VRAM when combined with gradient checkpointing.

8.6 Candidate Generation and MBR Consensus Decoding

Beam search maximizes the joint sequence likelihood,

$$\hat{\mathbf{y}}_{\text{beam}} = \arg\max_{\mathbf{y}} \prod_{t=1}^{T} P(y_t \mid y_{<t}, \mathbf{x}), \tag{9}$$

which in report generation tends to favor safe, high-frequency normal phrasing over less frequent but specific findings. To counter this, inference is formulated as MBR decoding [21] over a candidate pool $\mathcal{C} = \{\mathbf{y}_1, \ldots, \mathbf{y}_N\}$. The pool contains $N_b = 4$ beam-search candidates (repetition penalty 1.05, no repeated 3-grams, at most 128 new tokens) and $N_s = N - N_b$ stochastic samples generated with temperature $T = 0.70$ and top-$p = 0.90$. Candidates are truncated to 60 words, and empty or duplicate candidates are removed, so the effective pool can be smaller than N. The final implementation uses $N = 16$ (4 beams and 12 samples); the full-test evaluation reported in this paper uses $N = 8$ (4 beams and 4 samples), and $N = 16$ is evaluated on a 50-study pilot (Section 10). The consensus report is the candidate that maximizes the average utility against all other candidates:

$$\mathbf{y}^*_{\text{MBR}} = \arg\max_{\mathbf{y}_i \in \mathcal{C}} \frac{1}{|\mathcal{C}| - 1} \sum_{j \neq i} \mathcal{U}(\mathbf{y}_i, \mathbf{y}_j), \tag{10}$$

where the utility combines ROUGE-L with unigram and bigram F-measures:

$$\mathcal{U}(\mathbf{a}, \mathbf{b}) = 0.6 \cdot \text{ROUGE-L}_{\beta=1.2}(\mathbf{a}, \mathbf{b}) + 0.4 \cdot \frac{\text{F1}_{\text{uni}}(\mathbf{a}, \mathbf{b}) + \text{F1}_{\text{bi}}(\mathbf{a}, \mathbf{b})}{2}. \tag{11}$$

The pairwise computation uses Python longest-common-subsequence operations over short strings (≤ 60 words) and requires no Java. Because the medoid maximizes agreement with the rest of the pool, statements that occur in only a single rollout are disfavored; this is the verification role of the consensus stage. The selected report is evaluated with the official toolkit.

8.7 Empirical Baselines (E0 and E1)

Two zero-parameter baselines are scored on the locked test partition ($N = 589$) with the same tokenizer and scorer as the learned models.

8.7.1 E0: Constant Normal Report

The constant baseline outputs the same statement of normality for every study:

> "the lungs are clear. no focal consolidation, pneumothorax, or pleural effusion. the cardiac silhouette is normal."

A generator that does not exceed E0 has not learned more than a single static sentence applied to every patient.

8.7.2 E1: Pure Retrieval Top-1

An L2-normalized 1,536-dimensional study representation is formed from the two views:

$$\mathbf{v}_{\text{study}} = \left[\frac{\bar{\mathbf{z}}_{\text{front}}}{\|\bar{\mathbf{z}}_{\text{front}}\|_2} \; ; \; \frac{\bar{\mathbf{z}}_{\text{lat}}}{\|\bar{\mathbf{z}}_{\text{lat}}\|_2} \right]. \tag{12}$$

All 2,060 training studies are indexed in an exact inner-product FAISS index. For each test study j, the best training match

$$k^* = \arg \max_{i \in \mathcal{D}_{\text{train}}} \langle \mathbf{v}_{\text{test}}^{(j)}, \mathbf{v}_{\text{train}}^{(i)} \rangle \tag{13}$$

is found, and its report is output verbatim. This baseline measures how much score can be obtained from nearest-neighbor lookup alone.

8.8 End-to-End Inference Pipeline

During inference, the frontal and lateral radiographs of an unseen study are encoded by the frozen RAD-DINO encoder and pooled into 288 visual tokens. The 1,536-dimensional global vector is passed to the logistic-regression classifier to obtain finding probabilities, and the same vector queries the training-only FAISS index to retrieve the two most similar reports. The visual tokens are projected, the ChatML prompt is assembled, and the fine-tuned Qwen2.5-3B-Instruct model generates the candidate pool. MBR consensus decoding selects the final report.

9 Algorithm / Pseudo Code

9.1 Algorithm: Dual-View RAG-VLM Report Generation with MBR Decoding

Input: Frontal image I_f, lateral image I_l

Output: Generated findings report R^*

Offline preparation:

1. Select dual-view studies with non-empty findings; clean and truncate reports to 60 words; create and lock the study-level split (2,060 / 294 / 589).
2. Encode all images with the frozen RAD-DINO encoder, pool each view to 144 tokens, and cache the features.

3. Form 1,536-dimensional study vectors by concatenating the frontal and lateral vectors, L2-normalize them, and build a FAISS inner-product index over training studies only.

4. Fit the logistic-regression classifier; obtain 5-fold out-of-fold finding probabilities for training studies and classifier probabilities for validation and test studies.

5. Stage 1: train the projector (warm-up). Stage 2: train the projector, classification head, and LoRA adapters with $\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{CE}} + 0.2\,\mathcal{L}_{\text{BCE}}$.

Inference:

1. Load the pooled features of I_f and I_l and add view embeddings to obtain 288 visual tokens.

2. Compute the study vector $\mathbf{g}$ and predict the 14 finding probabilities $\hat{\mathbf{p}}$.

3. Retrieve the top-2 most similar reports from the training-only FAISS index.

4. Project the visual tokens and build the ChatML prompt from visual tokens, finding probabilities, and retrieved reports.

5. Generate a candidate pool of N reports (4 beams and $N-4$ samples with $T = 0.7$, $p = 0.9$); truncate to 60 words and remove empty and duplicate candidates.

6. For each candidate c_i, compute its mean utility $\frac{1}{|\mathcal{C}|-1} \sum_{j \neq i} \mathcal{U}(c_i, c_j)$.

7. Return the candidate with the highest mean utility as R^*.

9.2 End-to-End Processing Summary

Frontal + Lateral X-Ray → Frozen RAD-DINO → 1,536-d Vector + 288 Visual Tokens → Logistic Regression + FAISS Retrieval → MLP Projector + ChatML Prompt → Qwen2.5-3B (QLoRA) → Candidate Pool → MBR Consensus → Final Report

10 Experimental Setup and Results

10.1 Experimental Setup

10.1.1 Hardware and Hyperparameters

All experiments were run on consumer-grade GPU hardware, primarily an NVIDIA GeForce RTX 4050 Laptop GPU (6.00 GB VRAM, compute capability 8.9), with PyTorch 2.7.0 (CUDA 12.8) and bfloat16 precision. The configuration was:

- **Batch size:** effective batch size 16 (per-device batch size 1 with 16 gradient-accumulation steps on the 6 GB GPU).
- **Optimizer:** AdamW ($\beta_1 = 0.9$, $\beta_2 = 0.999$, weight decay 0.01).
- **Stage-2 learning rates:** LoRA 5×10^{-5} with linear warmup over 10% of the steps and cosine annealing down to 1×10^{-6}; projector 2×10^{-5}; classification head 1×10^{-4}.
- **Regularization:** LoRA dropout 0.10.
- **Maximum sequence length:** 640 tokens (288 visual tokens and 352 text and retrieval tokens).
- **Training seed:** 42 (single run).

10.1.2 Evaluation Toolkit

All headline scores are computed with the official `pycocoevalcap` toolkit using the Stanford PTBTokenizer: BLEU-1 to BLEU-4, METEOR 1.5, ROUGE-L, and CIDEr. References and outputs are lowercased and truncated to 60 words.

10.2 Headline Benchmark Evaluation

Table 4 compares the baselines with the decoding variants of the proposed model on the full test set ($N = 589$).

The final implementation uses a larger pool of $N = 16$ candidates. Its evaluation so far is limited to a 50-study pilot (Table 5). Because 50 studies is a small sample and the pilot was not run on the full test set, these values are not placed in Table 4 and should not be compared directly with it.

Table 4: Evaluation on the IU X-Ray test set ($N = 589$). All values are produced with `pycocoevalcap` and the PTBTokenizer.

System	BLEU-1	BLEU-2	BLEU-3	BLEU-4	METEOR	ROUGE-L	CIDEr
E0 Constant normal report	0.2390	0.1447	0.0932	0.0603	0.1455	0.2809	0.1855
E1 Pure retrieval (top-1)	0.3449	0.2020	0.1299	0.0871	0.1514	0.2513	0.1970
Dual-View RAG-VLM (beam search)	0.2728	0.1676	0.1155	0.0821	0.1360	0.2648	0.2915
Dual-View RAG-VLM (stochastic sampling)	0.3275	0.1980	0.1274	0.0840	0.1512	0.2753	0.2483
Dual-View RAG-VLM (MBR, $N = 8$)	0.3820	0.2355	0.1525	0.0975	0.1816	0.2951	0.3240

Table 5: Pilot evaluation of MBR decoding on a 50-study subset of the test set. These values are indicative only.

System	BLEU-1	BLEU-2	BLEU-3	BLEU-4	METEOR	ROUGE-L
MBR, $N = 16$	0.4053	0.2546	0.1666	0.1024	0.1926	0.3124

The $N = 16$ pilot CIDEr is 0.3026. An earlier $N = 8$ pilot on a 50-study subset gave BLEU-1 0.4007, METEOR 0.1875, and CIDEr 0.3812.

The pilot values for $N = 16$ and the earlier $N = 8$ pilot are close (BLEU-1 0.4053 versus 0.4007, METEOR 0.1926 versus 0.1875), and CIDEr is lower with $N = 16$ (0.3026 versus 0.3812). With 50 studies and no significance test, this does not establish that the larger pool is better, and a full-test comparison is left for future work.

For context, Table 6 lists scores reported in the original publications of two published systems. These systems use different training data and, in most cases, large-scale pre-training, so the comparison is indicative only.

Table 6: Scores reported in the original publications (not re-run under the present protocol).

Model (Venue)	BLEU-1	BLEU-4	METEOR	ROUGE-L
DART (CVPR 2025) [15]	0.486	0.208	0.205	0.411
KiUT (CVPR 2023) [16]	0.525	0.185	0.242	0.409
Dual-View RAG-VLM	0.382	0.098	0.182	0.295

10.3 Analysis of Decoding Strategies

1. **Beam search:** With four beams, the model reaches BLEU-1 of 0.2728. This is below the pure retrieval baseline on BLEU-1 to BLEU-4 and METEOR, and below the constant report on ROUGE-L (0.2648 versus 0.2809), although its CIDEr (0.2915) is higher than both baselines. Beam search tends to converge on generic normal phrasing, consistent with template mode collapse.

2. **Stochastic sampling:** Sampling ($T = 0.7$, $p = 0.9$) increases diversity and raises BLEU-1 to 0.3275 and METEOR to 0.1512, but the variance between samples

remains high. Sampled reports sometimes contain malformed or unsupported text (for example, an unsupported statement about asbestos exposure for one study, and a fixed phrase that repeats across several unrelated studies).

3. **MBR consensus:** Selecting the medoid of an eight-candidate pool raises BLEU-1 to 0.3820 (a 40.0% relative gain over beam search) and METEOR to 0.1816 (a 33.5% relative gain). MBR with $N = 8$ exceeds both E0 and E1 on every metric in Table 4.

10.4 Clinical Diagnostic Linear Probe

To check that the frozen dual-view representations encode clinically relevant information, a multi-label logistic probe was trained on the cached 1,536-dimensional features for the 14 CheXpert categories. This probe is the image classifier whose probabilities are used as priors. Table 7 reports the AUC for each finding; the macro-average AUC is 0.8135.

Table 7: AUC of the linear probe on frozen RAD-DINO dual-view representations.

Clinical finding	**AUC**
Pleural Effusion	0.9670
Pneumonia	0.9408
Cardiomegaly	0.8936
Pneumothorax	0.8688
Consolidation	0.8628
No Finding	0.8462
Support Devices	0.8277
Edema	0.8254
Lung Opacity	0.8241
Atelectasis	0.7975
Pleural Other	0.7790
Fracture	0.7634
Lung Lesion	0.6931
Enlarged Cardiomediastinum	0.5000*
Macro-average	**0.8135**

*An AUC of exactly 0.5 indicates no discrimination; this label may have too few positive cases in the evaluation split for a meaningful estimate.

Acute findings such as pleural effusion (0.967), pneumonia (0.941), and pneumothorax (0.869) are well separated by the frozen features before any language generation takes place, while lung lesion (0.693) and enlarged cardiomediastinum (0.500) remain difficult.

10.5 Qualitative Case Analysis

Two test studies compare stochastic sampling (full-test run) with MBR consensus decoding with $N = 16$ (50-study pilot run) and the radiologist findings. These examples are illustrative and do not by themselves establish a systematic effect.

Case 1: UID 1860 (thoracic spine deformity and degenerative change).

> **Ground truth:** the cardiac and mediastinal silhouettes are normal. the lungs are well-expanded and clear. there is no focal airspace opacity. there is no pneumothorax or effusion. there is dextrocurvature of the thoracic spine. there is deformity of the t9 vertebral body. levocurvature of the lumbar spine with significant degenerative change is also noted.
>
> **Stochastic sampling:** the heart size and mediastinal silhouette are within normal limits. there is no focal consolidation, pneumothorax or large pleural effusion. there are degenerative changes of the thoracic spine.
>
> **MBR consensus ($N = 16$):** the cardiomediastinal silhouette is normal in size and contour. atherosclerosis of the aorta. the lungs are clear bilaterally. there is no pneumothorax or large pleural effusion. there are no acute bony findings.

Both outputs agree with the reference on the normal heart and lungs. The sampled report mentions spinal degenerative changes, whereas the MBR report states that there are no acute bony findings and adds an aortic statement that is not in the reference. Neither describes the T9 deformity.

Case 2: UID 418 (stable cardiomegaly with prior granulomatous disease).

> **Ground truth:** stable enlargement of the cardiac silhouette, stable mediastinal and hilar contours, surgical clips and cabg markers. stable densities in the left base compatible with scarring or chronic subsegmental atelectasis. no focal alveolar consolidation, no definite pleural effusion seen. right hilar calcifications suggest a previous granulomatous process. no typical findings of pulmonary edema.
>
> **Stochastic sampling:** there is stable cardiomegaly. aorta is calcified and tortuous. there are multiple pleural calcifications representing prior asbestos exposure. these appear unchanged. there is no pneumothorax, pleural effusion, or focal airspace consolidation.
>
> **MBR consensus ($N = 16$):** the heart size and mediastinal contours are within normal limits. there is tortuous aorta. the lungs are clear bilaterally. no focal consolidation, pneumothorax or large pleural effusion.

The sampled report identifies the cardiomegaly but includes an unsupported statement about asbestos exposure. The MBR report removes that statement, but it also describes

the heart size as normal, contradicting the reference. This shows that consensus selection removes statements that occur in only a single rollout, but it can also discard a correct abnormal finding when most candidates describe a normal study.

Further observation. In the 50-study pilot, several studies whose reference describes an abnormality received a normal-template report from MBR. For example, UID 2370 (reference: consolidation and atelectasis in the right lower lobe with blunted right costophrenic angle) and UID 2485 (reference: bilateral lower lung airspace disease and small to moderate bilateral pleural effusions) both received reports describing clear lungs and no effusion. This tendency is consistent with the strength of the E0 and E1 baselines and is discussed in the next section.

11 Discussion and Limitations

11.1 Interpreting Scores on a Small, Repetitive Benchmark

The IU X-Ray collection is small (2,943 usable studies) and structurally uniform. A zero-parameter retrieval baseline reaches BLEU-1 of 0.3449, and a single constant sentence reaches ROUGE-L of 0.2809, so a substantial part of any model's n-gram score reflects the report template distribution rather than visual understanding. Reporting such baselines alongside learned models is therefore necessary for interpreting results on this dataset. The same reasoning suggests caution when comparing with systems that report much higher BLEU-1, since such systems also differ in pre-training data and evaluation details.

11.2 Semantic Metrics and Gap to Published Systems

Under MBR decoding with $N = 8$ the proposed model reaches METEOR of 0.1816, which is 0.0234 below the value reported for DART (0.2050), despite no MIMIC-CXR pre-training and consumer-grade hardware. On the other metrics the gap is larger: BLEU-1 of 0.3820 against 0.486 for DART and 0.525 for KiUT, BLEU-4 of 0.0975 against 0.208 and 0.185, and ROUGE-L of 0.2951 against 0.411 and 0.409. Because METEOR credits stem and synonym matches, a modern language-model backbone may benefit on this metric when it uses clinically equivalent wording. This explanation is a hypothesis; it has not been tested here, and the scores were obtained under different protocols.

11.3 Role of the Classifier and of Consensus Verification

The classifier gives an image-based estimate of 14 findings, and its out-of-fold probabilities are supplied to the generator as prompt indicators. The consensus stage then favors the candidate that agrees most with the rest of the pool. Together they are intended to ground the report and to suppress statements that appear in only one rollout, and the qualitative cases show that this does happen. However, the effect of the classifier priors alone was not isolated by an ablation, and agreement between report and classifier was not measured, so this paper does not claim that the reports are clinically verified. The classifier itself has a macro-AUC of 0.8135 and is near chance for some labels, so its probabilities can be wrong. A separate post-generation check of the report against the classifier output is a natural extension and is listed as future work.

11.4 Systems Feasibility

The combination of spatial pooling (144 tokens per projection), 4-bit NF4 quantization, LoRA ($r = 32$), and gradient checkpointing allows the pipeline to be trained with an effective batch size of 16 in about 4,921 MB of VRAM, within the 6 GB limit of the laptop GPU. This indicates that rigorous multimodal experiments on radiology data do not require large GPU clusters.

11.5 Limitations

- **Final configuration only partly evaluated:** The sixteen-candidate pool used by the final implementation was evaluated only on a 50-study pilot. The full-test results are for $N = 8$.
- **Normal-template tendency:** Even with consensus decoding, abnormal studies are often described as normal (Section 10), and the strong E0 and E1 baselines show that much of the achievable score on this dataset comes from normal-report wording.
- **Lexical overfitting:** Training on 2,060 reports without MIMIC-CXR pre-training limits generalization. LoRA dropout (0.10) and a reduced learning rate (5×10^{-5}) mitigate but do not remove this limitation.
- **Inference latency:** Generating eight candidates required about 23.7 seconds per study (about 0.04 studies per second) on the laptop GPU, and a sixteen-candidate pool costs more. This suits offline batch processing but not real-time use without parallel sampling or candidate pruning.

- **Incomplete ablation:** Only the baselines and the three decoding strategies were evaluated. The individual contributions of the lateral view, clinical priors, and retrieval were not isolated, for example through frontal-only, no-prior, and no-retrieval variants.
- **Clinical metrics and statistics:** Clinical-efficacy metrics (such as CheXbert F1 or RadGraph F1), multiple random seeds, and significance tests are not reported, so the evidence for clinical factuality is limited to n-gram metrics, the linear probe, and qualitative cases.
- **Protocol choices:** Study-level splitting cannot exclude patient overlap, because patient identifiers are unavailable. The probe AUC of exactly 0.5 for enlarged cardiomediastinum likely reflects too few positive cases.

12 Conclusion and Future Scope

12.1 Conclusion

This work presented the Dual-View RAG-VLM, a reproducible multimodal framework for chest X-ray report generation that is trained and evaluated on consumer-grade GPU hardware. Auditing the IU X-Ray benchmark showed that trivial baselines are strong: a constant normal report reaches ROUGE-L of 0.2809, and copying the top-1 retrieved report reaches BLEU-1 of 0.3449. With the same fine-tuned model, beam search remained below the retrieval baseline on BLEU-1 (0.2728) and tended toward generic templates, whereas MBR consensus decoding with eight candidates reached BLEU-1 of 0.3820, METEOR of 0.1816, and CIDEr of 0.3240 on the full test set without changing the model weights, exceeding both baselines on every metric. A 50-study pilot of the final sixteen-candidate configuration gave BLEU-1 of 0.4053 and METEOR of 0.1926, but this has not been confirmed on the full test set. A linear probe on the frozen dual-view features reached a macro-AUC of 0.8135, indicating that the visual representation carries meaningful clinical signal. The results are limited by the small dataset, the single training seed, the absence of large-scale pre-training, the normal-template tendency of the generated reports, and an incomplete ablation; they remain below published state-of-the-art scores and are not directly comparable with them.

12.2 Future Scope

1. **Full evaluation of the final configuration:** Evaluate the sixteen-candidate pool on all 589 test studies and compare it with $N = 8$.

2. **Complete ablation:** Evaluate frontal-only, no-prior, and no-retrieval variants, with multiple seeds and paired bootstrap tests, to isolate each component's contribution.

3. **Post-generation verification:** Add an explicit stage that compares the selected report with the classifier probabilities, flags or corrects contradicting statements, and measure its effect against radiologist references.

4. **Clinical evaluation:** Add clinical-efficacy metrics such as CheXbert F1 and RadGraph F1, an abnormal-only subset analysis, and evaluation by radiologists.

5. **External pre-training:** Pre-train or continue training on a larger chest X-ray report corpus such as MIMIC-CXR to raise the n-gram ceiling.

6. **Faster consensus decoding:** Parallel or speculative sampling and candidate pruning to reduce the per-study latency of MBR.

7. **Encoder adaptation:** Unfreeze the top blocks of the visual encoder with a small learning rate, at the cost of feature caching.

8. **Reinforcement learning:** Fine-tune with a reward combining ROUGE-L, METEOR, and a clinical-entity score, with KL and length regularization.

9. **Patient-level isolation and validation:** Use datasets with patient identifiers for patient-level splits, and undertake prospective clinical validation under appropriate ethical, privacy, and regulatory requirements.

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